

Project Description

Problem 1

i Project teammates

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ii Describe the collected data

Goal

We aim to examine whether the long-term relationship assumed in classical pairs trading remains stable over time and to investigate whether a time-varying hedge ratio (a dynamic beta calculated via the Kalman filter) can enhance the robustness and profitability of the pairs trading strategy.

Pairs

Main pair: **KO–PEP** (similar consumer beverage businesses).

Contrasting pair: **XOM–CVX** (oil majors; potentially more regime-sensitive).

Source and price

Daily **Adjusted Close** from **Yahoo Finance** to account for splits/dividends.

Sample window: 2015-01-01 to 2026-01-01 (daily).

Raw files (CSV)

- `KO_2015-01-01_2026-01-01.csv`
- `PEP_2015-01-01_2026-01-01.csv`
- `XOM_2015-01-01_2026-01-01.csv`
- `CVX_2015-01-01_2026-01-01.csv`

Each file contains Yahoo columns (`Date`, `Open`, `High`, `Low`, `Close`, `Adj Close`, `Volume`). We use `Date` and `Adj Close` in this project.

Construction

After aligning trading dates and dropping missing observations, let P_t^X, P_t^Y denote adjusted close prices in a pair. We use log prices $x_t = \log(P_t^X)$, $y_t = \log(P_t^Y)$ and log returns $r_t = \Delta \log(P_t)$. The static baseline uses OLS

$$y_t = \alpha + \beta x_t + \varepsilon_t, \quad s_t^{\text{static}} = y_t - \hat{\alpha} - \hat{\beta} x_t.$$

We also compute a rolling z-score (window $W = 252$)

$$z_t = \frac{s_t^{\text{static}} - \mu_t}{\sigma_t},$$

where μ_t, σ_t are rolling mean/std of s_t^{static} .

iii Descriptive statistics

We summarize (a) single-asset daily log returns and (b) pair-level diagnostics (static baseline).

Ticker	Mean	Std	Skew	Kurt	Min	Max
KO	-0.000309	0.011228	0.821762	10.343775	-0.062702	0.101709
PEP	-0.000271	0.012008	0.448434	18.800564	-0.121689	0.121322
XOM	-0.000265	0.017284	0.188605	6.456925	-0.119310	0.130511
CVX	-0.000280	0.018466	0.966415	25.795047	-0.205015	0.249906

Table 1: Daily log return summary (2015-01-01 to 2026-01-01).

Pair	$\text{Corr}(r^X, r^Y)$	$\hat{\beta}$	$\text{Std}(s^{\text{static}})$	% $ z > 2$
KO-PEP	0.725794	0.868284	0.095088	7.881
XOM-CVX	0.833666	0.963535	0.17081	3.651

Table 2: Pair-level diagnostics (static baseline).

iv Graphics

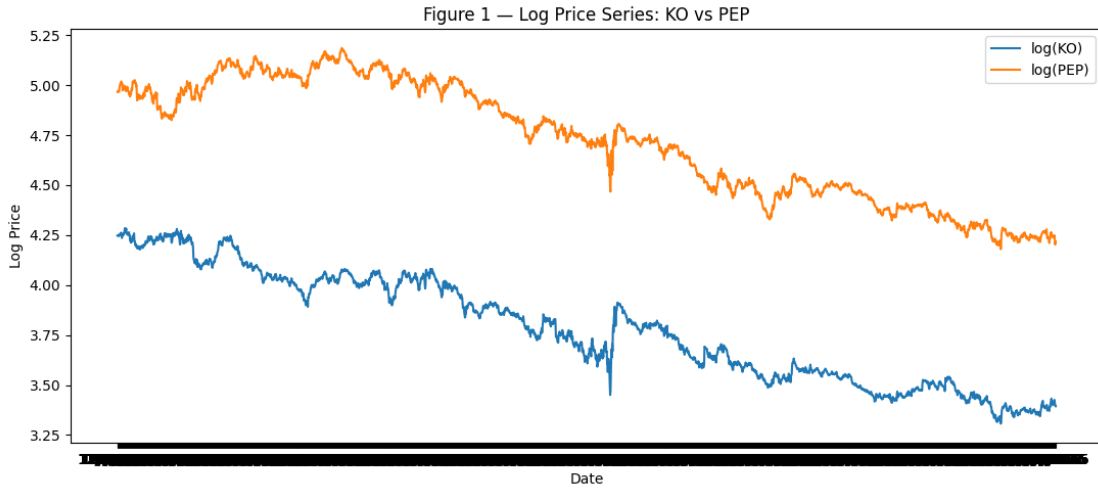


Figure 1: KO vs PEP: log adjusted close prices

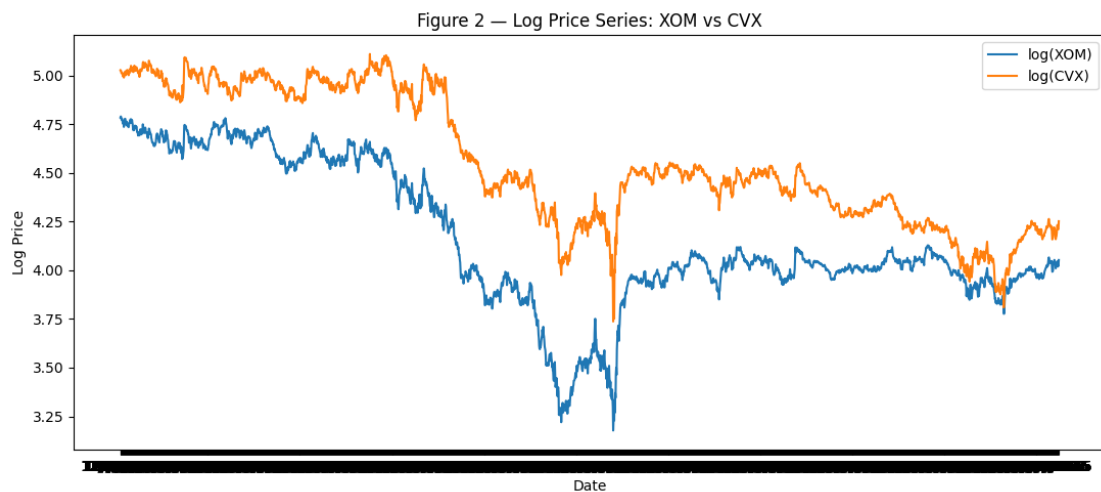


Figure 2: XOM vs CVX: log adjusted close prices.

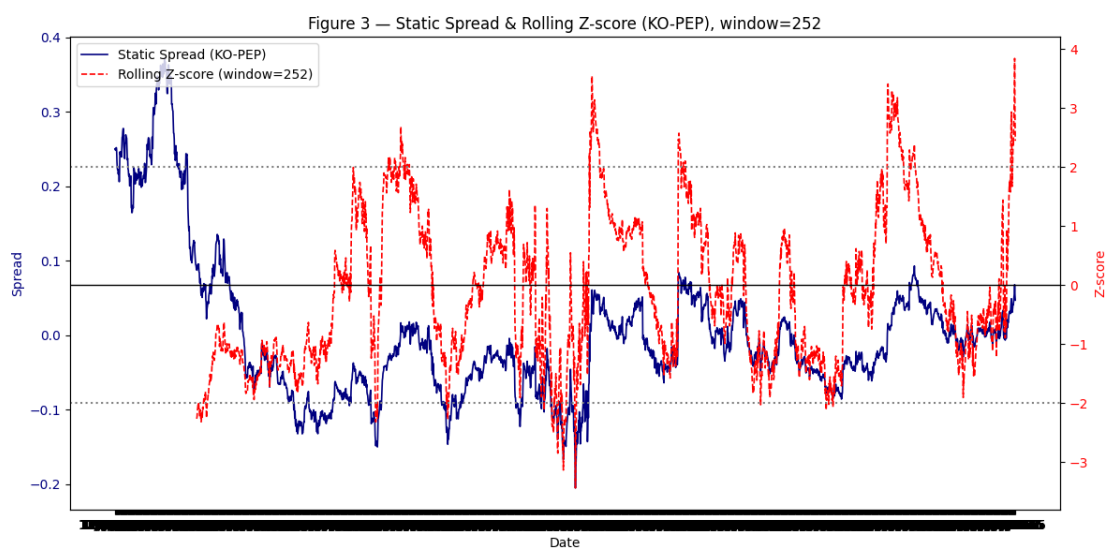


Figure 3: KO-PEP: static spread and rolling z-score ($W = 252$).

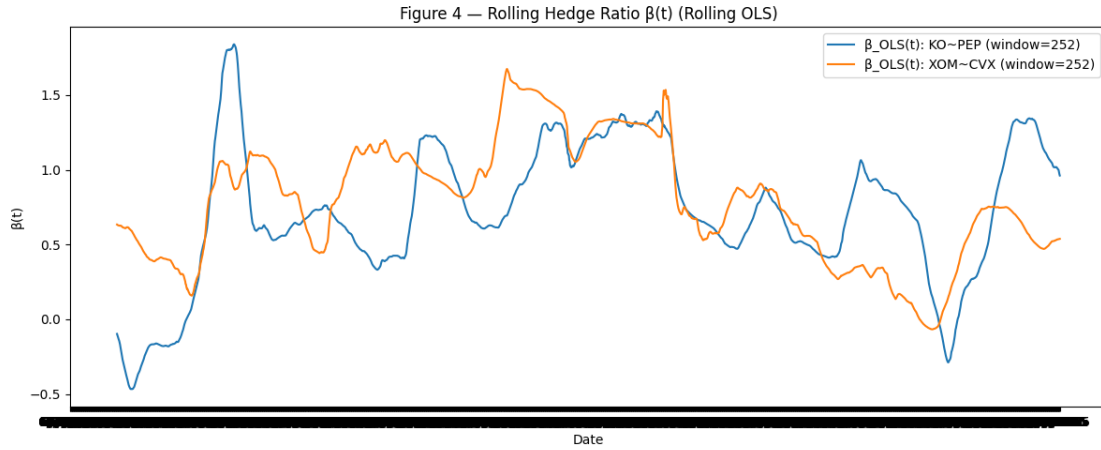


Figure 4: Rolling OLS hedge ratio $\hat{\beta}_{\text{roll}}(t)$ ($W = 252$) for both pairs.

Figure 5 — 3D: Time vs Z-score vs Rolling ADF p-value

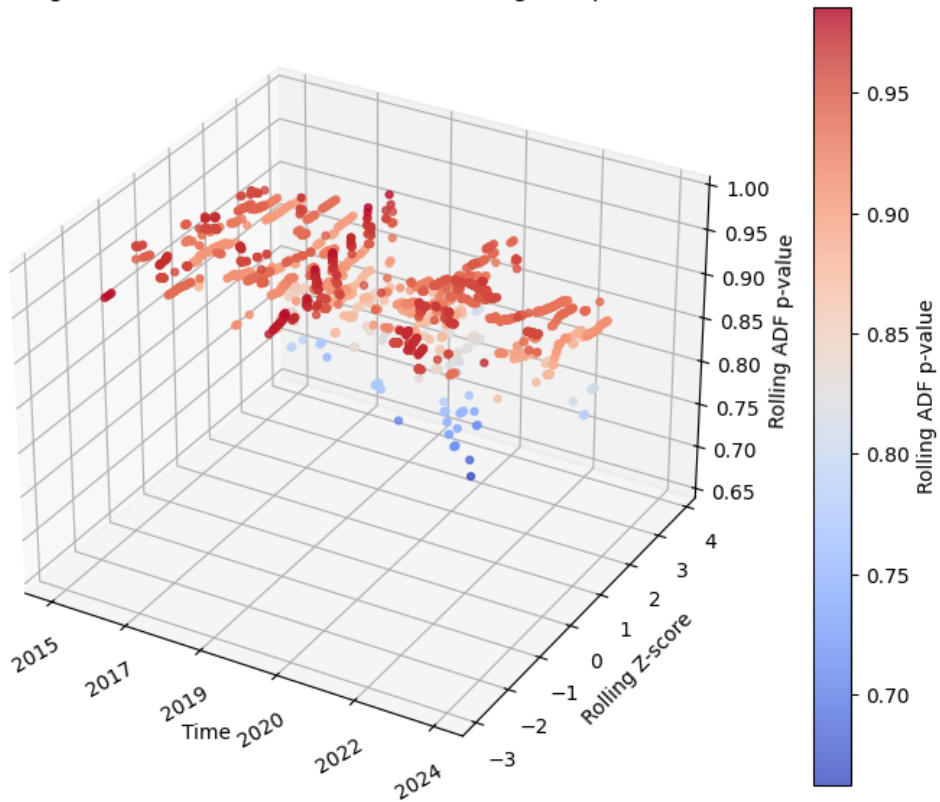


Figure 5: 3D plot: time (x), z-score (y), color = rolling ADF p-value of the spread.

v Comments

Table 1 shows clear cross-sector differences: KO/PEP have lower daily volatility than XOM/CVX, and all four return series exhibit strong fat tails (large excess kurtosis), implying that extreme moves are non-negligible and spread signals may be less stable under shocks.

Table 2 summarizes the static baseline. Both pairs have strong return comovement, with XOM–CVX more highly correlated than KO–PEP. However, the static spread is much more volatile for XOM–CVX, while KO–PEP exhibits a higher frequency of large standardized spread deviations ($|z| > 2$) under our normalization. Overall, high correlation alone does not guarantee a stable residual relationship under a fixed hedge ratio, motivating a time-varying hedge ratio model (state-space / Kalman filter) in the final project.

Figures 1–2 show strong co-movement within both pairs, but also clear divergence episodes (more pronounced for XOM–CVX), suggesting the long-run relationship may be regime-dependent. Figure 3 shows that during the observation period for KO–PEP, the standardized spread signal (rolling z-score) exhibits clustered extreme values and time-varying amplitude, rather than a stable mean-reverting pattern. Figure 4 further indicates that the hedge ratio is not constant: rolling OLS estimates drift materially for both pairs, so a fixed- β specification can be misspecified in some periods and may itself induce spread instability. Figure 5 links spread deviations to rolling stationarity diagnostics: periods with large $|z|$ tend to coincide with weaker evidence of stationarity (higher rolling ADF p-values), reinforcing the hypothesis that “tradability” varies over time. Overall, the graphics motivate our final-project extension to a time-varying hedge ratio model (state-space / Kalman filter) and a regime-aware evaluation of spread stability.

vi Research Questions

This project addresses four key research questions:

1. whether the selected stock pairs exhibit statistically significant and stable cointegration;
2. whether the constant hedge ratio assumption underlying traditional pairs trading holds over time;
3. whether incorporating a time-varying hedge ratio via a Kalman filter improves spread stationarity and trading robustness; and
4. whether mean reversion strength is regime-dependent across different market conditions.

The objective of this project is to conduct a comprehensive statistical evaluation of pairs trading strategies by integrating cointegration analysis with time-varying parameter modeling. First, we aim to test whether selected stock pairs within the same industry (KO–PEP and XOM–CVX) exhibit a stable long-run equilibrium relationship using Engle–Granger cointegration tests. We then investigate whether this equilibrium relationship remains constant over time or experiences structural drift across different market regimes.

A key research question is whether the traditional assumption of a constant hedge ratio is appropriate. To address this, we compare a static OLS-based hedge ratio with a time-varying hedge ratio estimated via a Kalman filter in a state-space framework. We examine how dynamic hedge

ratio estimation affects spread stationarity, z-score behavior, and the persistence of deviations from equilibrium.

Additionally, we evaluate whether mean reversion strength is regime-dependent by analyzing rolling ADF p-values and rolling hedge ratios. This allows us to assess whether periods of weakened stationarity correspond to prolonged spread deviations and reduced tradability.

Finally, we compare the performance of static versus dynamic trading rules based on spread z-scores, including entry/exit behavior and sensitivity to structural shifts. Overall, the project seeks to determine whether incorporating time-varying parameters enhances the robustness and adaptability of statistical arbitrage strategies under changing market conditions.